

3. (a) (i) The diagram below shows the position of the Earth's continents today.



In 1912 Alfred Wegener suggested that all the continents must once have been joined together as one big land mass.

Diagrams **A**, **B** and **C** show the position of the Earth's continents 50 million, 100 million and 150 million years ago, but not necessarily in that order.



A



B



C

Give the letter, **A**, **B** or **C**, of the diagram which shows the position of the Earth's continents 150 million years ago. [1]

Letter



- (ii) Wegener’s theory of continental drift was not accepted by other scientists until several years after his death in 1930. The evidence to support his theory was found in 1960 when part of the ocean floor was surveyed around a plate boundary. The table shows data collected from the survey.

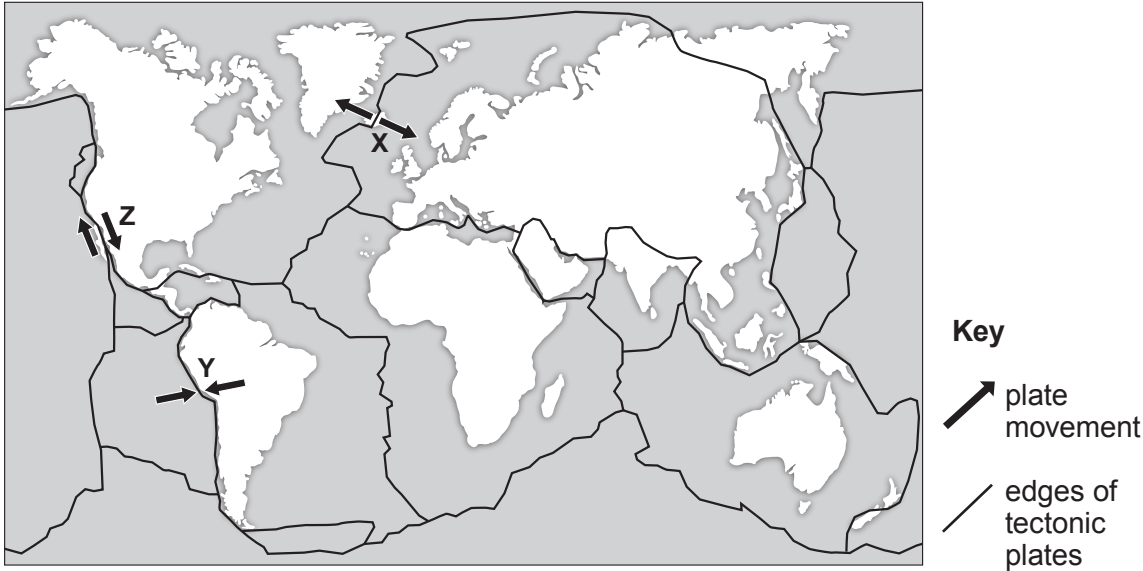
Distance of ocean floor from plate boundary (km)	Approximate age of rock (million years)
2000	100

Calculate the mean speed at which the ocean floor is spreading. [1]

mean speed (km/million years) = $\frac{\text{distance (km)}}{\text{time (million years)}}$

Mean speed = km/million years

- (iii) The map shows some information about tectonic plates and three locations X, Y and Z.



Give the **letter** of the location you would expect to have earthquakes but not volcanic eruptions. [1]

Letter

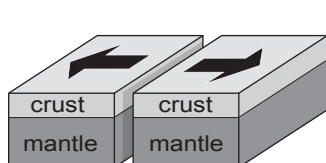
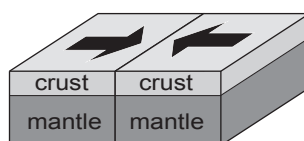
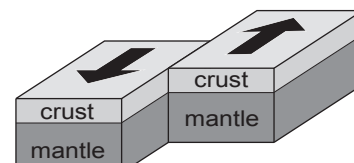


- (b) The photograph below shows 'pillow lava' which was formed from volcanoes on the sea bed at a **constructive** plate boundary millions of years ago.



'pillow lava' on Llanddwyn Island, Anglesey

- (i) Tick (✓) the box of the diagram that shows a constructive plate boundary where the pillow lava was formed. [1]


☐

☐

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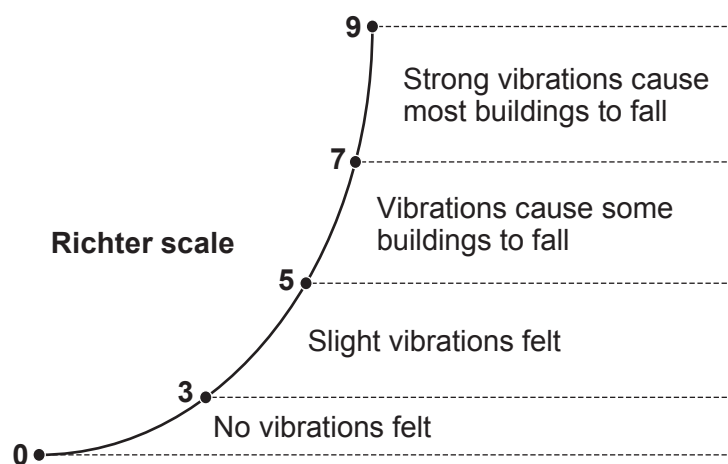
- (ii) Complete the sentences by underlining the correct word(s) in the brackets. [2]

Pillow lava is formed at a constructive plate boundary when
(**magma** / **sea water** / **crust**) rises and cools, forming new rock.

The movement of the Earth's tectonic plates is caused by
(**electric currents** / **convection currents** / **ocean currents**) within the mantle.



- (c) Charles Richter developed the Richter Scale in 1935 to measure the strength of earthquakes.



In June 2018 an earthquake occurred in the Caernarfon area, with a minor tremor being felt.

Circle the number that best shows the size of the earthquake in Caernarfon. [1]

1 4 6 8

3410U101
11

7

